STALKWISE INJECTIVITY OF H^M(R LIM K_N) -> H^M(K_{N(X)})

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Sheafification computes cohomology. Cohomology of Sheaves, Lemma cohomology-lemma-sheafification-cohomology. H^m(R lim K_n) is the sheafification of U |-> H^m(U, R lim K_n).
- Milnor sequence. Cohomology of Sheaves, Lemma cohomology-lemma-RGamma-commutes-with-Rlim. The short exact sequence for H^m(U, R lim K_n).

Source reference. Original lemma 0BKJ: sources/stacks-chow/tags.txt, source lines 14562–14562. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D60: sources/stacks-chow/tags.txt, source lines 16505–16505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let (K_n) be an inverse system in $D(\mathcal{O}_X)$. Let $x \in X$ and $m \in \mathbf{Z}$. Assume there exist an integer $n(x)$ and a fundamental system $\mathfrak{U}_x$ of open neighbourhoods of x such that for $U \in \mathfrak{U}_x$*

(1) *$R^1 \lim H^{m-1}(U, K_n) = 0$, and*
(2) *$H^m(U, K_n) \to H^m(U, K_{n(x)})$ is injective for $n \geq n(x)$.*

Then the map on stalks $H^m(R \lim K_n)_x \to H^m(K_{n(x)})_x$ is injective.

Proof. Let γ be an element of $H^m(R \lim K_n)_x$ which maps to zero in $H^m(K_{n(x)})_x$. Since $H^m(R \lim K_n)$ is the sheafification of $U \mapsto H^m(U, R \lim K_n)$ (by Lemma 0BKJ) we can choose $U \in \mathfrak{U}_x$ and an element $\tilde{\gamma} \in H^m(U, R \lim K_n)$ mapping to γ. Then $\tilde{\gamma}$ maps to $\tilde{\gamma}_{n(x)} \in H^m(U, K_{n(x)})$. Using that $H^m(K_{n(x)})$ is the sheafification of $U \mapsto H^m(U, K_{n(x)})$ (by Lemma 0BKJ again) we see that after shrinking U we may assume that $\tilde{\gamma}_{n(x)} = 0$. For this U we consider the short exact sequence

$$0 \to R^1 \lim H^{m-1}(U, K_n) \to H^m(U, R \lim K_n) \to \lim H^m(U, K_n) \to 0$$

of Lemma 0D60. By assumption (1) the group on the left is zero and by assumption (2) the group on the right maps injectively into $H^m(U, K_{n(x)})$. We conclude $\tilde{\gamma} = 0$ and hence $\gamma = 0$ as desired. □

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